

Ali Baigelenov

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<http://abaigele.github.io>

EDUCATION

Ph.D. in Technology – Purdue University | May 2026

Advisor: Dr. Paul Parsons

Dissertation: “Inspiration in Visualization Design: Perception, Discourse, and Epistemic Mediation”

<https://doi.org/10.25394/PGS.32098147>

M.S. in Computer Graphics Technology – Purdue University | December 2017

Advisor: Dr. Paul Parsons

Thesis: “Interactivity Factors in Visualization-Based Exploratory Search”

B.S. in Computer Graphics Technology – Purdue University | December 2015

PROFESSIONAL APPOINTMENTS

Senior Lecturer - SDU University, Kazakhstan | January 2024 - Present

School of Information Technology and Applied Mathematics

Department of Computer Science

RESEARCH INTERESTS

information visualization; human-computer interaction; human-AI interaction; design cognition;
human-centered design; applied cognition

PUBLICATIONS

My name appears in bold. All publications listed are peer-reviewed.

Full papers

1. Parsons, P., Agrawal, S., Bui, P., Shukla, P., and **Baigelenov, A.** (Accepted, to appear at DRS 2026). Bias Reconsidered: Ecologically Rational Heuristics in Design Cognition. In *Proceedings of the DRS 2026 Design Research Society 55th Conference*
2. **Baigelenov, A.**, Shukla, P., Bui, P., and Parsons, P. (2026). Talking Inspiration: A Discourse Analysis of Data Visualization Podcasts. In *CHI Conference on Human Factors in Computing*

Systems (CHI '26), April 13-17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 11 pages.
<https://doi.org/10.1145/3772318.3791537>

3. **Baigelenov, A.**, Shukla, P., and Parsons, P. (2025). How Visualization Designers Perceive and Use Inspiration. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26-May 1, 2025, Yokohama, Japan. ACM, New York, NY, USA, 12 pages.
<https://doi.org/10.1145/3706598.3714191>

Short papers and Extended abstracts

1. **Baigelenov, A.**, Shukla, P., Zhang, Z., and Parsons, P. (2025). Are Cognitive Biases as Important as they Seem for Data Visualization?. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26-May 1, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3719926>
2. Shukla, P., Bui, P., Levy, S., Kowalski, M., **Baigelenov, A.**, and Parsons, P. (2025). De-skilling, Cognitive Offloading, and Misplaced Responsibilities: Potential Ironies of AI-Assisted Design. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26-May 1, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages.
<https://doi.org/10.1145/3706599.3719931>
4. Parsons, P., Gray, C. M., **Baigelenov, A.**, & Carr, I. (2020). Design judgment in data visualization practice. In *2020 IEEE Visualization Conference (VIS)* (pp. 176-180). IEEE.
5. Parsons, P., **Baigelenov, A.**, Hung, Y. H., & Schrank, C. (2020). What Design Methods do DataVis Practitioners Know and Use?. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1-8).
6. **Baigelenov, A.**, & Parsons, P. (2018). Interactivity Factors in Visualization-Based Exploratory Search. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems (CHI EA '18)*. ACM, New York, NY, USA, Paper LBW009, 6 pages. DOI: <https://doi.org/10.1145/3170427.3188558>

Workshop and Symposium papers

1. Parsons, P. C., Shukla, P. C., **Baigelenov, A.**, & Gray, C. M. (2023). Developing Framing Judgment Ability: Student Perceptions from a Graduate UX Design Program. In *Proceedings of the 5th Annual Symposium on HCI Education* (pp. 23-32).
2. Saenz, M., **Baigelenov, A.**, Hung, Y. H., & Parsons, P. (2017). Reexamining the cognitive utility of 3D visualizations using augmented reality holograms. In *Workshop on Immersive Analytics: Exploring Future Interaction and Visualization Technologies for Data Analytics (Immersive 2017)*, pp. 5.

Poster abstracts

1. Parsons, P., Hung, Y. H., & **Baigelenov, A.** (2018). Toward an Analysis of Practitioner-Oriented Resources for Visualization Design. In *IEEE VIS '18: Proceedings of the 2018 IEEE Conference on Information Visualization*, Poster Abstracts, pp. 2
2. **Baigelenov, A.**, Saenz, M., Hung, Y. H., & Parsons, P. (2017). Toward an Understanding of Observational Advantages in Information Visualization. In *IEEE VIS '17: Proceedings of the 2017 IEEE Conference on Information Visualization*, Poster Abstracts, pp. 2.

Full papers - Under review

1. **Baigelenov, A.**, Bui, P., Agrawal, S., Shukla, P., and Parsons, P. (Under Review). Metaphors We Inspire By: Inspiration and Imitation in Visualization Design.
2. Parsons, P., Shukla, P., Bui, P., Agrawal, S., and **Baigelenov, A.** (Under Review). Situatedness in Visualization Design: Making Unresolved Work Actionable.
3. Parsons, P., Gray, C., and **Baigelenov, A.** (Under Review). Design Knowledge in Data Visualization: Mapping the Epistemic Landscape

Short papers - Under review

1. **Baigelenov, A.** and Parsons, P. (Under Review). Beyond Visual Examples: Inspiration and Fixation Across Data, Code, and Visual Form.

PROFESSIONAL SERVICE

Organizing Committee

Workshop on Visualization for Communication (VisComm), IEEE VIS - 2025

Reviewer

ACM Conference on Human Factors in Computing Systems (ACM CHI) - 2019-2026

IEEE Visualization Conference (IEEE VIS) - 2023

IEEE Conference on Visual Analytics Science and Technology (IEEE VAST) - 2019

ACM Designing Interactive Systems Conference (ACM DIS) - 2025 (Special Recognition)

ACM Creativity and Cognition Conference (ACM C&C) - 2025 (Highly Useful Review)

Campus/Department

Quality Assurance Committee

SDU University, Kazakhstan

ABET Accreditation Preparation Committee

SDU University, Kazakhstan

Updated May 2026

Undergraduate Diploma Project Defense Committee
SDU University, Kazakhstan

AWARDS AND GRANTS

National

Kazakhstan's Governmental Scholarship "Bolashak" | January 2010

Institutional

Sriver Award | August 2018

Daniel and Martina Lewis Graduate Award | August 2018

TALKS AND POSTERS

National/International

1. "Talking Inspiration: A Discourse Analysis of Data Visualization Podcasts." *CHI Conference on Human Factors in Computing Systems*, Barcelona, Spain, 2026
2. "How Visualization Designers Perceive and Use Inspiration." *CHI Conference on Human Factors in Computing Systems*, Yokohama, Japan, 2025
3. Gateway Usability SGX3 Focus Week. *Science Gateways 2023 Annual Conference*. San Diego Supercomputer Center, San Diego, California, 2023

Campus/Departmental

1. Cognitive Tools: A Data-Driven Perspective. *Future Work and Learning*. Purdue University, Indiana, 2019

TEACHING

Instructor Positions, SDU University, Kazakhstan

- CSS 489: Diploma Preparation (2024, 2025)
- INF 232: Web Programming: Back End (2024, 2025, 2026)
- CSS 153: Computer Ethics: Legal, Ethical, and Societal Impacts of Computing (2024)
- CSS 410: Research Tools and Methods (2024)
- INF 228: UX/UI Design (2024)
- CSS 206: Database Management Systems I (2024, 2025)

Teaching Assistant Positions, Purdue University, IN, USA

- CGT 35300: Principles of Interactive and Dynamic Media (2021, 2022)
- CGT 35600: Web Programming, Development, and Data Integration (2021, 2022, 2023)
- CGT 45600: Advanced Web Programming, Development, and Data Integration (2022, 2023)

Curricular Innovations

- New Courses Developed or Co-Developed
 - CSS 153: Computer Ethics: Legal, Ethical, and Societal Impacts of Computing
 - INF 228: UX/UI Design

Teaching Interests

- human-computer interaction; human-AI interaction; human-centered design; information and data visualization; user experience design; user interface design; ethics in computing; web development front and backend; database management systems

MENTORSHIP AND SUPERVISION

Meirzhan Uristemov

Teaching Assistant, Master's Student, SDU University | March 2024 - June 2024

Merey Zhursinadilova

Undergraduate Research Assistant, SDU University | August 2025 - December 2025